Question 3

<table>
<tr><th>Question</th><th>Answers</th><th>Extra information</th><th>Mark</th><th>AO / Spec. Ref.</th></tr>
<tr><td>03.1</td><td>
<table>
<tr><th>Hazard</th><th>Risk</th><th>Plan to minimise risk</th></tr>
<tr><td>Iodine solution is an irritant</td><td>May cause allergic reaction or skin rash</td><td>wash skin immediately (after contact)
or
wear gloves
or
clean up spills

allow method to prevent spills e.g. use a dropper bottle

ignore do not spill</td></tr>
<tr><td>Sharp knife</td><td>may cut you / someone / skin</td><td>cut away from the body
or
cut on a chopping board
or
keep fingers away from blade (when cutting)

allow description of how to carry knife safely

ignore use a blunt knife</td></tr>
</table>
1 mark for each correct row</td><td>1

1</td><td>AO3
4.1.1.5
RPA1</td></tr>
</table>

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	**thin layer** (to) help see individual cells	allow so light can penetrate	1	AO2 4.1.1.5 RPA1
	iodine solution (to) stain / see the parts of the cell	allow visible named sub-cellular structures e.g. nucleus, cytoplasm, cell wall, starch grains ignore chloroplast ignore (to) stain the cell	1	
	at an angle (to) prevent / reduce air bubbles		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	*recall of equation* magnification = $\frac{\text{size of image}}{\text{size of real object}}$	allow magnification = $\frac{\text{length of image}}{\text{length of real object}}$ ignore use of equation triangle	1	AO1
	rearrangement of equation size of real object = $\frac{\text{size of image}}{\text{magnification}}$	allow length of real object = $\frac{\text{length of image}}{\text{magnification}}$ allow recall and rearrangement of equation implied at any stage	1	AO2
	substitution $\frac{4.8}{400}$	allow substitution of incorrectly converted value	1	AO2
	0.012 (cm)	allow answer using incorrectly converted value	1	AO2
	conversion 120 (μm)	allow conversion to μm at any stage	1	AO2 4.1.1.5 RPA1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	any **two** from: • include magnification / scale • use continuous lines **or** ensure no gaps in lines • do not draw overlapping cells • draw (wider) cell walls • do not shade • draw all the cells present • draw correct cell shapes • do not have gaps between cells • draw nuclei in correct location • label cell part(s)	ignore make it neater allow do not colour allow label named cell part(s)	2	AO3 4.1.1.2 RPA1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.5	(would) look more magnified / bigger	ignore reference to zoom	1	AO1 4.1.1.5
	(cell would) have more detail **or** (would) be at a higher resolution **or** (could) see more sub-cellular structures **or** sub-cellular structures seen in detail	allow correct examples of sub-cellular structures such as ribosomes, mitochondria, cell membrane ignore chloroplast allow (could) be in 3D allow would be in black and white	1	

Total Question 3		**14**